EXPERIMENT 3A: MODEL PLANNING AND BUILDING – CLASSIFICATION

LEARNING OBJECTIVES:

- Build classification models to predict customer churn.
- Split the dataset into training and testing sets.
- Train Logistic Regression, Decision Tree, and Random Forest classifiers.
- Evaluate the models using Accuracy, Precision, Recall, F1-Score, and Confusion Matrix.
- Compare the performance of the three classification models using a bar chart.
- Identify the model with the highest evaluation performance based on the selected metrics.
- Provide business recommendations to reduce customer churn.

METHODS / PYTHON LIBRARIES USED:

Library / Method	Purpose	Reason to Use
train_test_split	Split data into 80% training and 20% testing	Used to train the model and evaluate it on unseen data
LogisticRegression	Build a linear classification model	Simple and effective for binary classification
DecisionTreeClassifier	Build a tree-based classification model	Easy to understand and handles different types of features
RandomForestClassifier	Build an ensemble classification model	Combines multiple decision trees to improve prediction
accuracy_score	Measure overall prediction accuracy	Shows the percentage of correctly classified customers

PROGRAM CODE:

```
import import pandas as pd
import matplotlib.pyplot as plt
```

```
import seaborn as sns

from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.tree import DecisionTreeClassifier
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score, precision_score, recall_score,
f1_score, confusion_matrix

df = pd.read_csv("CustomerChurn_Preprocessed.csv")

X = df.drop("Churn", axis=1)
y = df["Churn"]

X = pd.get_dummies(X, drop_first=True)

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)



    print("Precision:", round(pr, 3))
    print("Recall   :", round(r, 3))
    print("F1 Score :", round(f, 3))
    print("Confusion Matrix:\n", confusion_matrix(y_test, p))

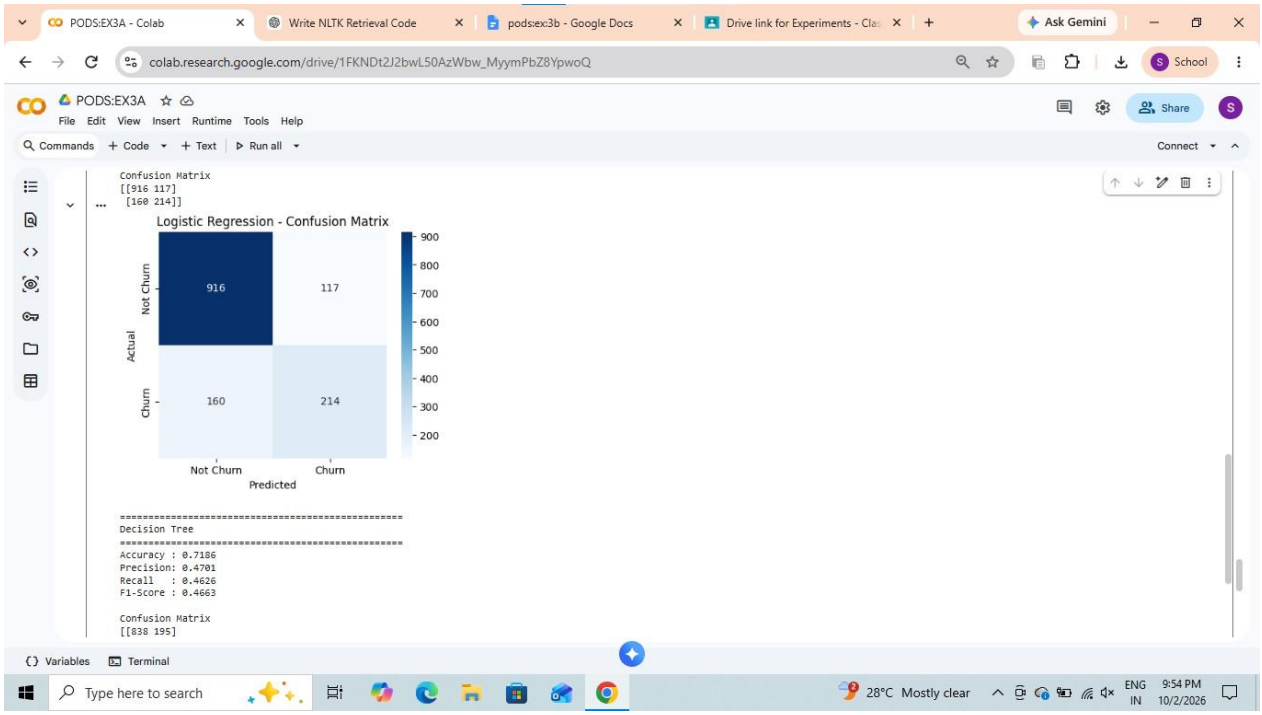
result = pd.DataFrame(
    scores,
    columns=["Model", "Accuracy", "Precision", "Recall", "F1"]
)

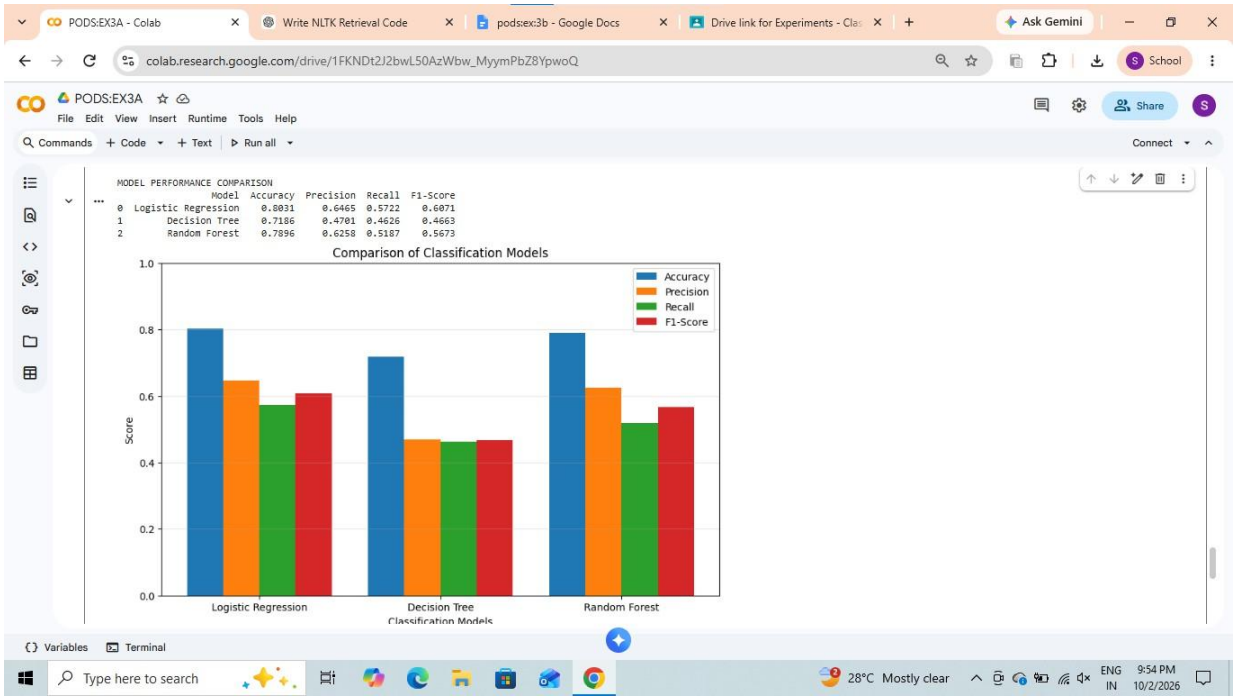
print("\nModel Comparison:")
print(result)

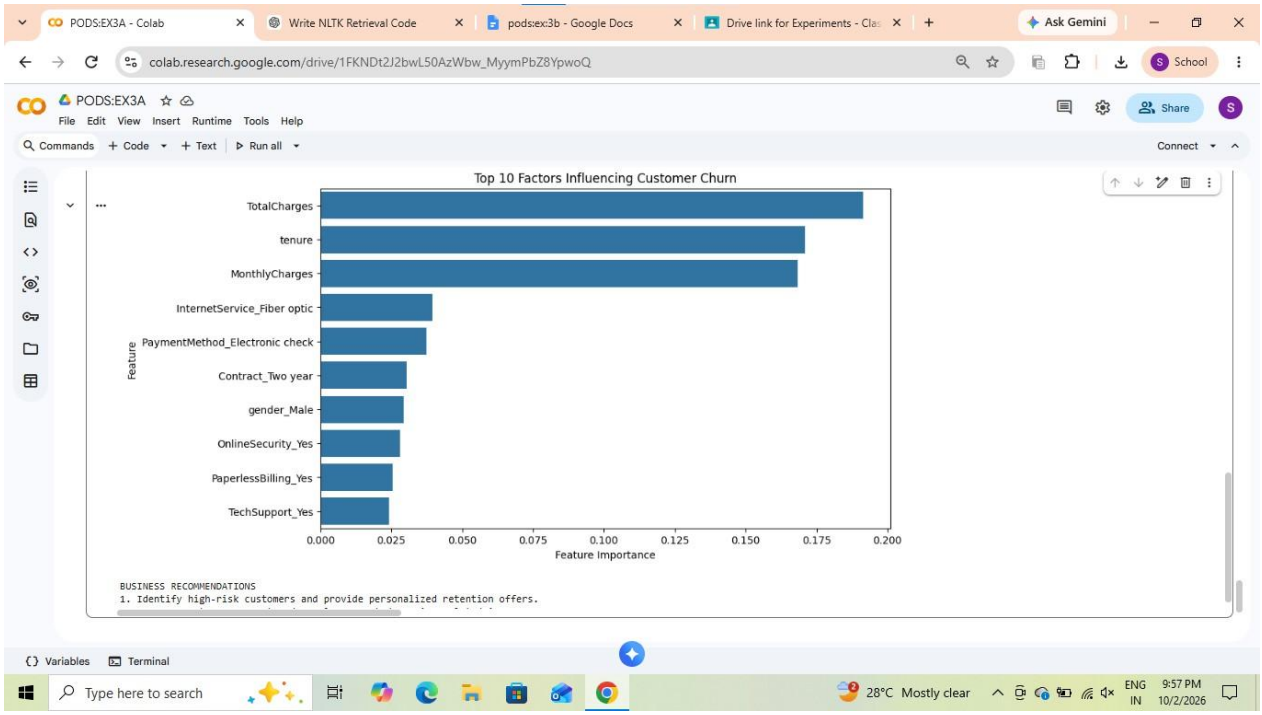
result.set_index("Model").plot(kind="bar", ylim=(0, 1))
plt.title("Model Performance Comparison")
plt.ylabel("Score")
plt.xticks(rotation=0)
plt.show()
```

```
print("\nBest Model:")
print(result.loc[result["F1"].idxmax(), "Model"])
```

OUTPUT:







LEARNING OUTCOME:

- Build classification models for customer churn prediction.
- Apply **Logistic Regression, Decision Tree, and Random Forest** algorithms.
- Evaluate models using **Accuracy, Precision, Recall, F1-Score, and Confusion Matrix**.
- Compare the performance of different classification models.
- Identify important factors related to customer churn.
- Provide suitable business recommendations to reduce customer churn.